

Solvency Field Theory V11.2 Toy Program Addendum

Retarded Dent Fields, Two-Slit Structure, Which-Way Tagging, and Erasure

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Status. This document is a toy-model addendum to *SFT V11.2 Seed Note: The Retarded Dent Field Program*. It does not claim theorem closure, a derivation of the Born rule, a solution to Bell’s theorem, or a completed quantum theory. Its purpose is narrower: to record a controlled sequence of numerical toys showing that the retarded-dent picture has dynamical content and produces the right qualitative pattern in several quantum-adjacent setups.

1 Purpose of the addendum

The V11.2 seed note proposed a research program:

$$\psi(\mathbf{x}, t) = \bar{\psi}(\mathbf{x}, t) + \delta\psi_{\text{ret}}(\mathbf{x}, t), \quad (1)$$

where $\bar{\psi}$ is the modeled lag field and $\delta\psi_{\text{ret}}$ is the unresolved, retarded dent history created by passing interactions and boundary-accounting structure.

The hard gate remains:

$$\delta\psi_{\text{ret}} \stackrel{?}{\Rightarrow} Q[\rho] = -\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}}, \quad (2)$$

with associated recovery of the Born rule, uncertainty bounds, Bell correlations, and no-signaling. The toy program does not pass that gate. It asks whether the visual mechanism has enough structure to be worth carrying forward.

2 The common toy geometry

The refined toy replaces an external random force bath with a geometric picture: a particle rides inside a deformed internal lag shell, the “inside-out sphere” of the V11.2 note. In two dimensions this is modeled as a dented circular equipotential shell:

$$U(r, \theta, t) = \frac{k}{2} [r - R(\theta, t)]^2. \quad (3)$$

The shell radius is deformed by angular modes and, later, by transient localized crossings:

$$R(\theta, t) = R_0 + \sum_n a_n(t) \cos(n\theta + \phi_n(t)) + R_{\text{transient}}(\theta, t). \quad (4)$$

The particle has continuous motion plus an internal tick phase:

$$\dot{\phi} = \omega_{\text{tick}}. \quad (5)$$

At tick events, the particle samples the local bowl gradient and receives a small impulse. In schematic form,

$$\Delta \mathbf{p}_{\text{tick}} \propto -\nabla U(r, \theta, t). \quad (6)$$

This is not a quantum postulate. It is the minimal mathematical version of the “motorcycle in the dented sphere” picture: the internal clock samples the local geometry, and the geometry changes the trajectory.

3 Toy 1: random dents alone are not enough

The first toy treated unresolved dents as an external colored force bath in a harmonic lag bowl. That model spread and heated the ensemble, but did not settle into a quantum-like stationary uncertainty scale.

Case	Δx	Δp	$\Delta x \Delta p$	Energy
Clean bowl	0.0397	0.0399	0.00158	0.00158
Retarded dents, no damping	0.375	0.377	0.142	0.142
Balanced bath reference	0.974	0.972	0.947	0.947

Interpretation. Dents treated as generic unresolved kicks produce diffusion/heating. The model needs geometry and backreaction, not noise alone.

4 Toy 1B: dented-bowl geometry creates stable trajectory spread

The second toy used the dented shell geometry directly. Three cases were compared: perfect bowl, static dents, and dynamic dents.

Case	Spatial spread	Momentum spread	Product	Radial spread	Energy
Perfect bowl	1.006	0.019	0.019	0.0085	0.425
Static dents	0.997	0.158	0.157	0.067	0.720
Dynamic dents	1.008	0.156	0.157	0.057	0.652

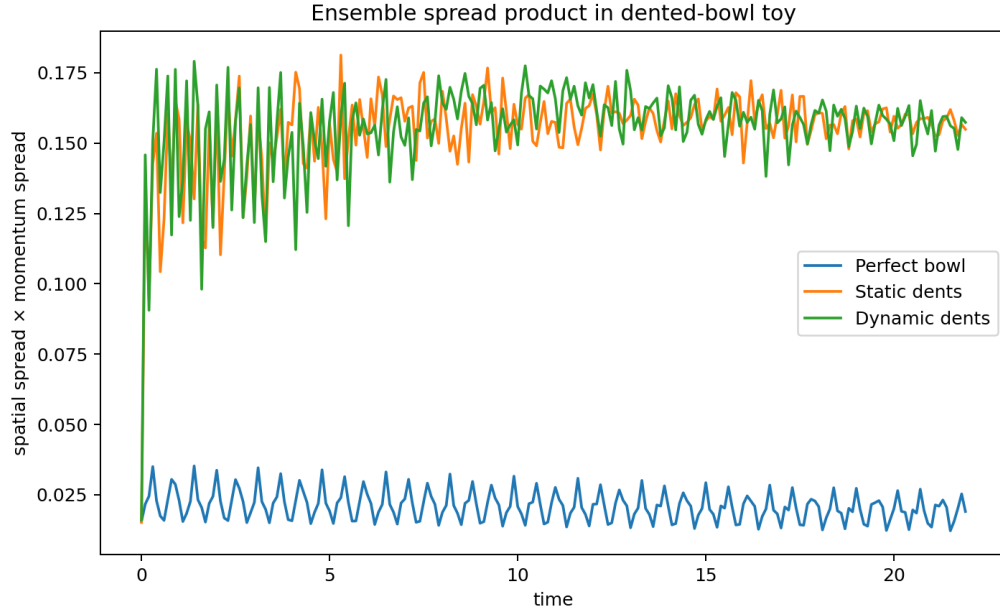


Figure 1: Denting the internal lag shell produces a stable, geometry-induced spread larger than the perfect bowl case.

Interpretation. The dented-bowl model produces structured spread without becoming a pure noise bath. The result supports the qualitative claim that deformation of the internal lag shell can redirect motion deterministically while appearing unpredictable at ensemble scale.

5 Parameter scan: what controls the spread?

A parameter scan varied dent amplitude, tick rate, bowl stiffness, and damping/backreaction proxy.

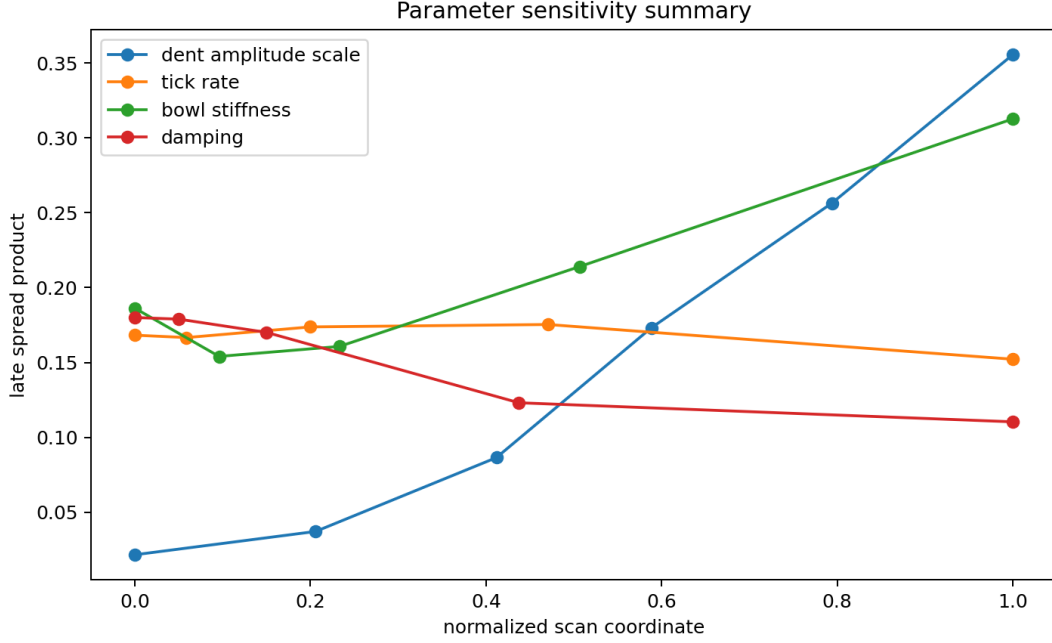


Figure 2: Parameter sensitivity summary. Dent amplitude and bowl stiffness dominate the spread. Damping/backreaction suppresses it. Tick rate has a weaker effect in this first tick rule.

Main lesson. In the first refined toy, ensemble spread is primarily geometric. It is controlled more by dent amplitude and shell stiffness than by the tick rate. This suggests that future toys should make the tick sample local curvature/normal direction more explicitly if tick rate is expected to dominate.

6 Toy 1C: photon-like crossings and shifting fields

The next toy added localized transient dents moving rapidly around the shell, plus an optional slow background drift. This models the idea that the bowl is continually reshaped by fast passing interactions.

Case	Product	Radial spread	Energy	Pair divergence growth
Dynamic dents only	0.1645	0.0573	0.6758	909.6×
Dynamic dents + photon crossings	0.2311	0.0638	0.8104	646.7×
Photon crossings + shifting field	0.2588	0.0657	0.8421	934.5×

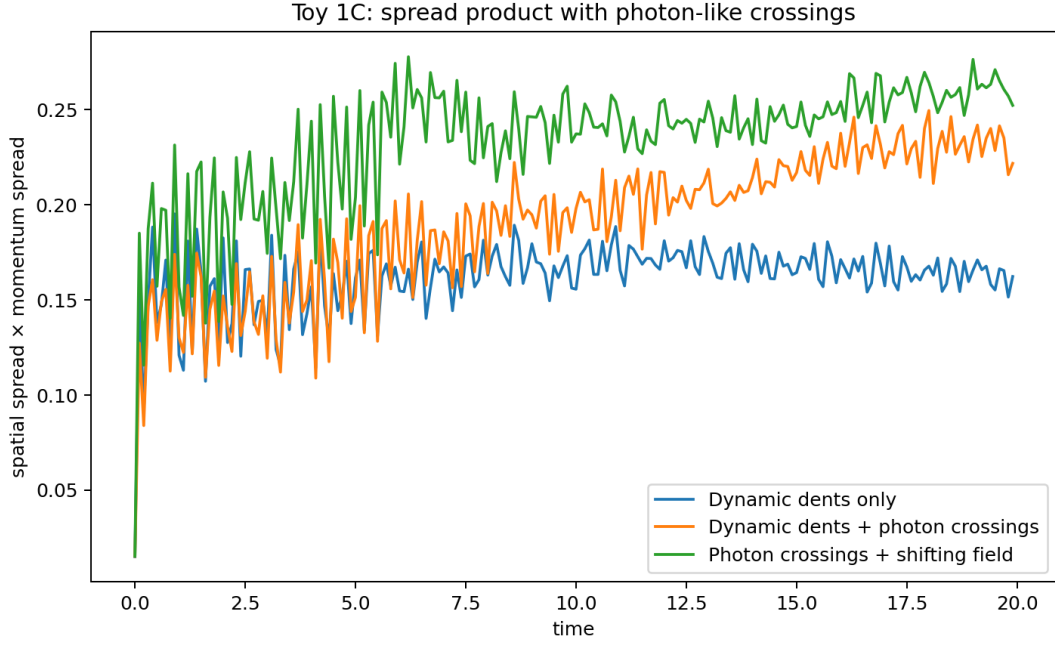


Figure 3: Photon-like crossings and slow drift increase the stable spread while preserving bowl-constrained structure.

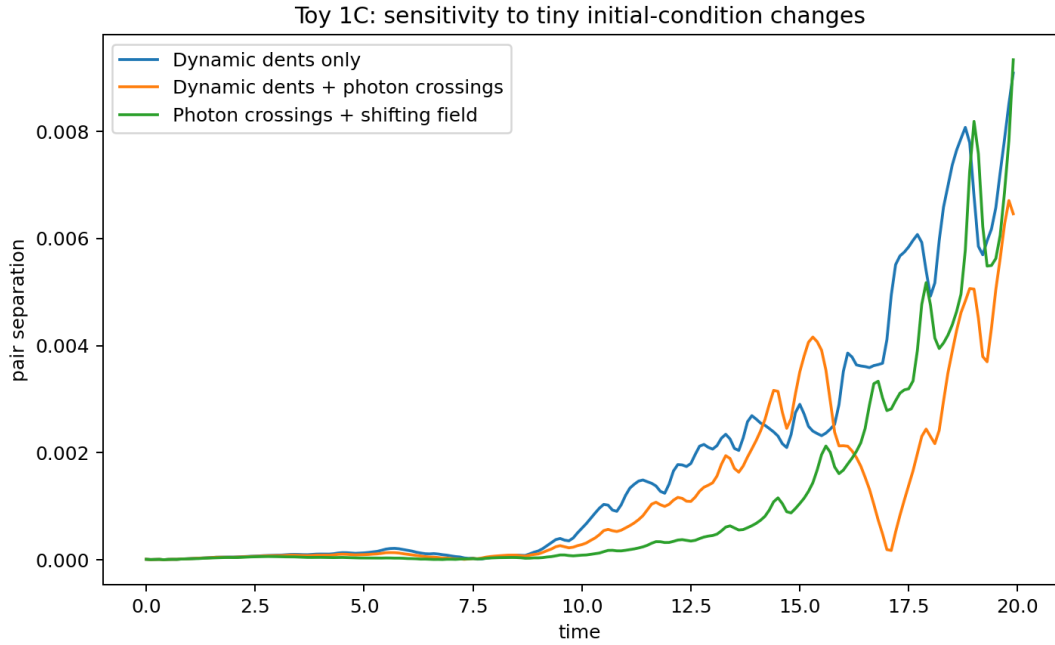


Figure 4: Tiny initial-condition differences grow strongly under the same dent history, illustrating practical unpredictability without inserting fundamental randomness.

Interpretation. Fast transient dents increase the spread and sensitivity while keeping the motion geometry-based. This supports the picture of deterministic but practically unreconstructable dent histories.

7 Toy 2: two-slit correlated dent field

The two-slit toy compared three cases: aperture only, local slit dents, and a correlated two-slit dent term. The correlated term was a schematic path-difference field:

$$F_y^{\text{corr}} \propto e^{-x/\lambda} \sin[\beta(r_1 - r_2)] \frac{\partial(r_1 - r_2)}{\partial y}. \quad (7)$$

This is not imported quantum amplitude. It is a classical dent-field proxy for a two-path geometric correlation.

Case	Reached screen	Screen width σ_y	Fringe-like peaks
Aperture only	423	0.2099	2
Local slit dents	431	0.1880	2
Tuned correlated dents	434	0.5255	10

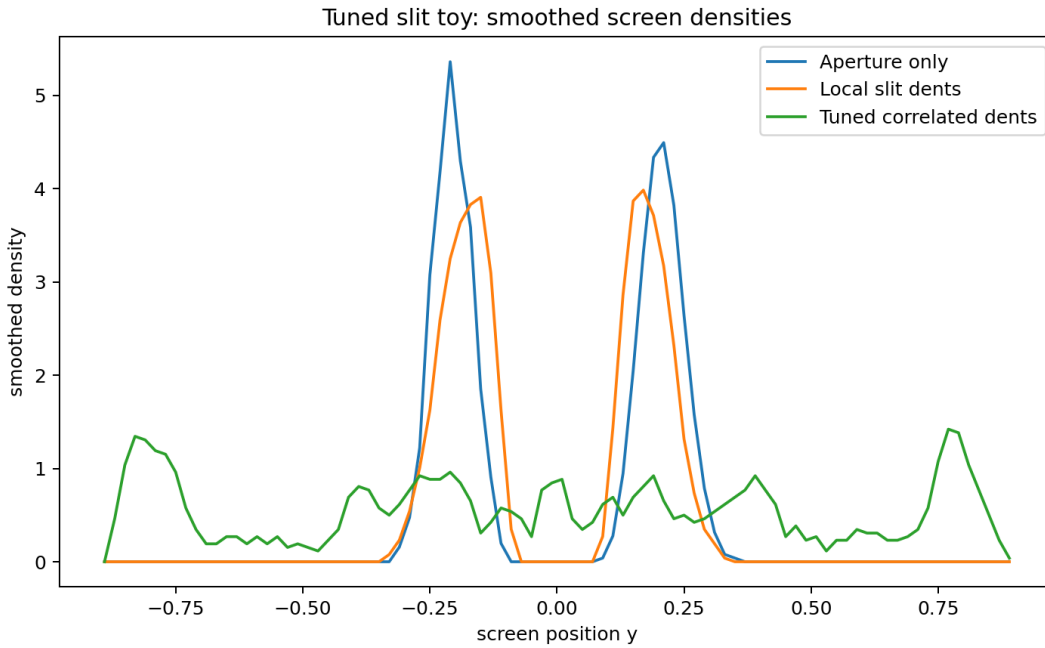


Figure 5: A tuned correlated two-slit dent term produces multi-peak screen structure beyond the aperture-only and local-dent cases.

Main lesson. Local dents alone did not generate fringe structure. Correlated two-path dent geometry did. This is a key qualitative result of the toy program.

8 Toy 3: which-way tagging versus physical kicks

The which-way detector was split into two independent knobs:

$$s_\phi = \text{path/phase tagging strength}, \quad s_p = \text{physical momentum-kick strength}. \quad (8)$$

This separates information tagging from brute-force scattering.

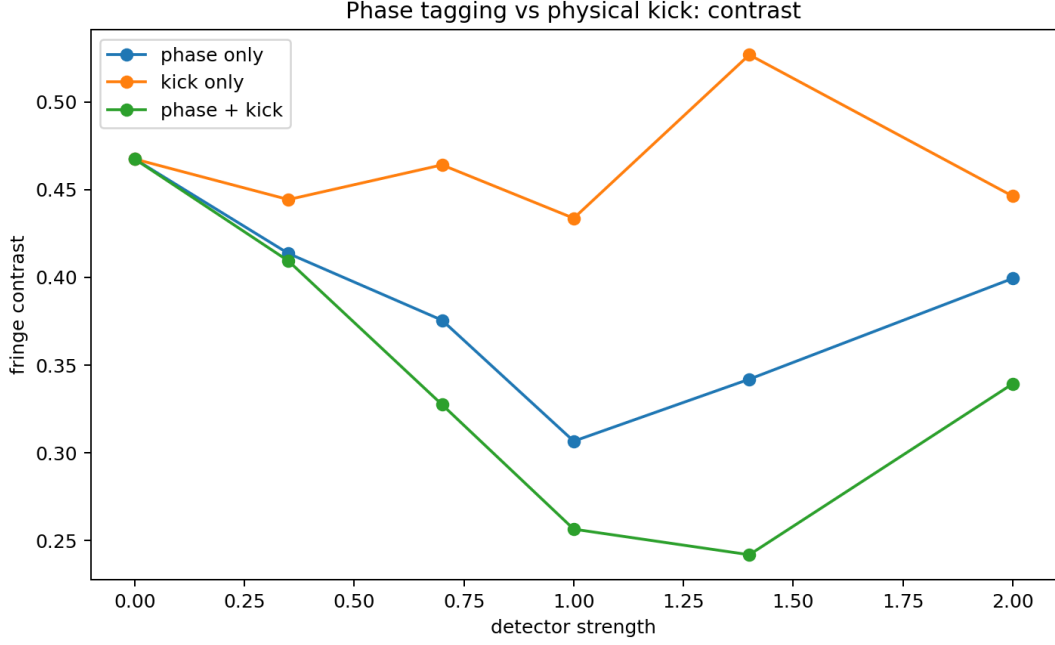


Figure 6: Phase tagging suppresses fringe contrast more cleanly than physical kicks. Kick-only disturbance creates new scattering structure rather than clean decoherence.

Interpretation. In the toy, which-way information is represented by scrambling the correlated two-path dent phase. Physical detector kicks are different: at high strength they deflect the particle and write their own scattering pattern. This reproduces the qualitative distinction between decoherence and direct disturbance.

9 Toy 4: quantum eraser analogue

The final toy introduced an eraser region downstream of the detector. The eraser resets the phase/path tag to a common value but cannot undo any physical kick. Five cases were compared.

Case	σ_y	Peaks	Contrast	Eraser hit
Coherent correlated dents	0.536	6	0.499	0.000
Phase tag only	0.540	10	0.270	0.000
Phase tag + eraser	0.520	6	0.459	0.112
Phase + kick + eraser	0.514	9	0.409	0.112
Kick only	0.522	12	0.458	0.000

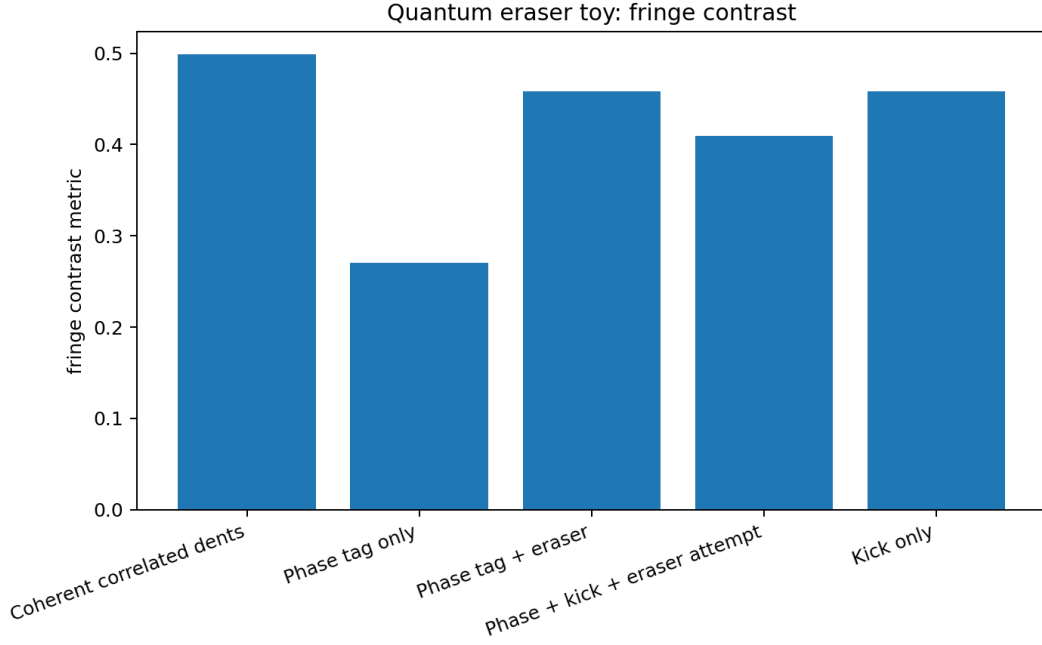


Figure 7: Phase tagging suppresses contrast. Erasing the phase tag restores much of it. If a physical kick is also present, the recovery is partial because the eraser cannot undo real deflection.

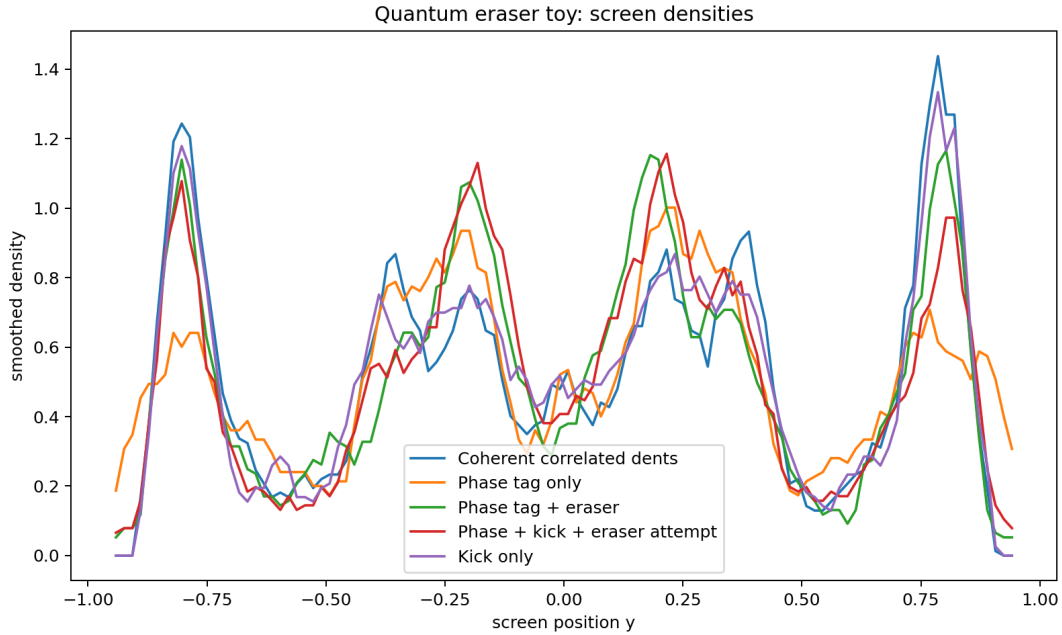


Figure 8: Selected screen densities from the eraser toy. The phase-tag-only case loses contrast; the phase-tag-plus-eraser case recovers much of the coherent structure.

Main result. The eraser toy produced the desired qualitative sequence:

$$\text{coherence} \rightarrow \text{phase tag suppresses structure} \rightarrow \text{phase eraser restores structure.} \quad (9)$$

When a kick is present, recovery is incomplete because path-information erasure is not the same as undoing

physical scattering.

10 What the toy program now supports

The toy program supports the following limited claims:

1. A dented internal lag shell produces structured trajectory spread, not merely random heating.
2. Photon-like transient crossings and slow background drift amplify unpredictability while preserving geometric motion.
3. Local slit dents alone do not produce a fringe train, but correlated two-path dent geometry can.
4. Which-way phase tagging suppresses correlated screen structure more cleanly than physical momentum kicks.
5. A downstream eraser that removes the phase/path tag can restore much of the contrast, while physical kicks leave residual disturbance.

11 What remains unproved

The toy program does not yet establish quantum mechanics. The hard gates remain:

$$\delta\psi_{\text{ret}} \xrightarrow{?} Q[\rho] = -\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}}, \quad (10)$$

$$P(x) = |\Psi(x)|^2, \quad (11)$$

$$\Delta x \Delta p \geq \frac{\hbar}{2}, \quad (12)$$

plus Bell-violating correlations with no superluminal signaling.

Correct status. The Retarded Dent Field Program remains a speculative quantum-foundations extension of SFT. The toy sequence justifies further work because it reproduces several qualitative signatures - spreading, two-path structure, which-way suppression, and eraser-like recovery - using one coherent dent-field grammar. It does not replace the Born rule, Schroedinger evolution, Bell analysis, or no-signaling proofs.

12 Next mathematical gates

The next version of the program should stop adding visual toys and aim at the first formal reduction. Three gates are recommended.

Gate 1: derive an effective quantum potential

The unresolved dent covariance should be measured and related to the ensemble density:

$$\langle \delta\psi(\mathbf{x}, t) \delta\psi(\mathbf{x}', t') \rangle = C(\mathbf{x} - \mathbf{x}', t - t'). \quad (13)$$

The goal is to show whether some limit of this covariance generates

$$Q[\rho] = -\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}}. \quad (14)$$

Gate 2: calibrate the uncertainty scale

The toy currently generates dimensionless spreads. A physical version must produce a natural scale,

$$\Delta x \Delta p \rightarrow \frac{\hbar}{2}, \quad (15)$$

without hand tuning the detector or dent amplitudes.

Gate 3: separate boundary nonlocality from signaling

If SFT treats the hidden variables as boundary-encoded dent state rather than local particle variables, the program must eventually show two things at once:

$$\text{Bell-type correlations are recoverable,} \quad (16)$$

while

$$\text{controllable superluminal signaling remains impossible.} \quad (17)$$

13 Conclusion

The toy program did not prove the quantum extension. It did something smaller but important: it showed that the inside-out sphere picture can be given equations, that dented lag geometry creates stable spread, that correlated two-path dents can generate interference-like structure, and that path tagging and erasure behave qualitatively differently from direct physical kicks.

The update to the V11.2 note is therefore simple:

Retarded dent fields are not yet a quantum theory, but they are now a viable toy program.

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